Ex. No.: 4a

Experiment Title: EMPLOYEE AVERAGE PAY

Aim:

To find out the average pay of all employees whose salary is more than 6000 and number of days worked is more than 4.

Algorithm:

1. Create a flat file named `emp.dat` with employee details including name, salary per day, and number of days worked.
2. Write an AWK script named `emp.awk`.
3. For each employee record:
   a. Check if the salary per day is greater than 6000 and the number of days worked is more than 4.
   b. If yes, print the employee's name and total pay (salary per day × days worked).
   c. Keep track of total employees satisfying the condition and their combined pay.
4. Finally, print the number of employees and the average pay of those employees.

Program Code:

emp.dat

```
JOE 8000 5
RAM 6000 5
TIM 5000 6
BEN 7000 7
AMY 6500 6
```

emp.awk

```
BEGIN {
    total_pay = 0
    count = 0
    print "EMPLOYEES DETAILS"
}
{
    salary = $2 * $3
    if ($2 > 6000 && $3 > 4) {
        print $1, salary
        total_pay += salary
        count += 1
    }
}
END {
    print "no of employees are= ", count
    print "total pay= ", total_pay
    if (count > 0)
        print "average pay= ", total_pay / count
    else
        print "average pay= 0"
}
```

Sample Input and Output:

```
[student@localhost ~]$ vi emp.dat
[student@localhost ~]$ vi emp.awk
[student@localhost ~]$ gawk -f emp.awk emp.dat
```

Output:

```
EMPLOYEES DETAILS
JOE 40000
BEN 49000
AMY 39000
no of employees are= 3
total pay= 128000
```

```
average pay= 42666.7
```

Result:

The program executed successfully. It displayed the employees who satisfy the condition and calculated their total and average pay correctly.

Ex. No.: 4b

Experiment Title: RESULTS OF EXAMINATION

Aim:
To print the pass/fail status of a student in a class based on their marks.

Algorithm:

1. Read the data from a file (`marks.dat`).

2. For each student, extract the marks from each column.

3. Compare the marks for all subjects:
   a. If any subject mark is less than 45, print "Fail" for that student.
   b. If all subject marks are 45 or more, print "Pass".

Program Code:

marks.dat

```
BEN 40 55 66 77 55 77

TOM 60 67 84 92 90 60

RAM 90 95 84 87 56 70

JIM 60 70 65 78 90 87
```

marks.awk

```
BEGIN {

    print "NAME SUB-1 SUB-2 SUB-3 SUB-4 SUB-5 SUB-6 STATUS"

    print
"-------------------------------------------------------------"

}

{
```

```
    status = "PASS"
    for (i = 2; i <= NF; i++) {
        if ($i < 45) {
            status = "FAIL"
            break
        }
    }
    print $1, $2, $3, $4, $5, $6, $7, status
}
END {
    print
"------------------------------------------------------------"
}
```

Sample Input and Output:

```
[root@localhost student]# gawk -f marks.awk marks.dat
```

Output:

```
NAME SUB-1 SUB-2 SUB-3 SUB-4 SUB-5 SUB-6 STATUS
------------------------------------------------------------
BEN 40 55 66 77 55 77 FAIL
TOM 60 67 84 92 90 60 PASS
RAM 90 95 84 87 56 70 PASS
JIM 60 70 65 78 90 87 PASS
```

Result:

The program executed successfully, displaying the pass/fail status for each student based on their marks.